```
import pandas as pd

url = "https://raw.githubusercontent.com/stedy/Machine-Learning-with-R-datasets/master/insurance.c
df = pd.read_csv(url)

df.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

Next steps: Generate code with df New interactive sheet

```
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1338 non-null   int64
 1   sex       1338 non-null   object
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64
 4   smoker    1338 non-null   object
 5   region    1338 non-null   object
 6   charges   1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

	age	bmi	children	charges
count	1338.000000	1338.000000	1338.000000	1338.000000
mean	39.207025	30.663397	1.094918	13270.422265
std	14.049960	6.098187	1.205493	12110.011237
min	18.000000	15.960000	0.000000	1121.873900
25%	27.000000	26.296250	0.000000	4740.287150
50%	39.000000	30.400000	1.000000	9382.033000
75%	51.000000	34.693750	2.000000	16639.912515
max	64.000000	53.130000	5.000000	63770.428010

```
df = pd.get_dummies(df, drop_first=True)
df.head()
```

	age	bmi	children	charges	sex_male	smoker_yes	region_northwest	region_southeast	re
0	19	27.900	0	16884.92400	False	True	False	False	
1	18	33.770	1	1725.55230	True	False	False	True	
2	28	33.000	3	4449.46200	True	False	False	True	
3	33	22.705	0	21984.47061	True	False	True	False	
4	32	28.880	0	3866.85520	True	False	True	False	

Next steps: Generate code with df New interactive sheet

```
from sklearn.model_selection import train_test_split

X = df.drop("charges", axis=1)
y = df["charges"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)
```

```
▾ LinearRegression
LinearRegression()
```

```
y_pred = model.predict(X_test)
y_pred
```

```
array([ 8969.55027444,  7068.74744287, 36858.41091155,  9454.67850053,
       26973.17345656, 10864.11316424,   170.28084136, 16903.45028662,
        1092.43093614, 11218.34318352, 28101.68455267,  9377.73460205,
        5263.0595179 , 38416.04221107, 40255.82339284, 37098.25353123,
       15240.39392306, 35912.88264434,  9112.52398703, 31461.92108909,
        3847.68845883, 10130.12001517,  2370.54189389,  7140.21550828,
       11301.76782638, 12961.65366224, 14509.47251876,  6159.8976107 ,
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        4561.82376202,  3408.20756033,  4459.81359745, 13032.06505076,
        1979.99357292,  8813.28303302, 33271.29124448, 32585.51583927,
        3908.76090964,  4326.10774721, 14142.81326533, 11423.45494846,
        8774.13955311, 12097.28051001,  5281.57353499,  3150.5596042 ,
       35494.46461214,  9150.1124786 , 15836.84575621,  2343.57470069,
       12364.78414194,  1482.29488266, 13389.06105161, 12573.57395972,
        4341.83680558, 32165.33688042, 13321.3360032 , 12896.82071102,
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       11839.25019431,  4061.19750503, 26652.40230125, 10930.14138671,
        2137.41385988,  6209.01123411, 10729.82391284, 11628.3104129 ,
       10981.04528946,  9166.50818596, 11954.27732874,  6747.85121734,
        7248.5304713 , 10735.16710748,  6580.84819774,  8762.00329355,
        3767.13383454, 36632.4975496 ,  6378.11979721, 30842.09248656,
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        9942.30149778, 14473.5260648 , 17693.37304474, 35258.24845137,
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       29444.04271523,  3674.48498404, 28308.26432106,  5823.36495229,
        5407.76752001,  1883.4947576 , 11499.675042  , 15075.90690632,
       11699.63163008,  4308.82427855,  9895.1840044 , 31708.40056201,
```

```
  -86.87094667, 32819.71429004,  3280.69178415, 10183.88853878,
14318.76389179, 31642.35684542, 11461.57806791,  3929.23701831,
13107.89313088, 31810.99450607,  8152.02593593,  3238.08417076,
 8439.56108376, 10594.63871458, 15219.68736374,  5647.8808143 ,
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12450.92458882,  4813.53293089,  2047.65528159, 32112.11251984,
25111.52085938, 17484.27663755, 26411.46181822, 10159.52421   ,
37260.32666386,  -441.23918333,  6779.55013103,  7781.45337795,
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15191.08806502, 11132.09935114,  6932.80116584,  2525.64793222,
 1536.05183213, 31944.78284317, 16414.12251517, 12011.53367195,
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25674.40250332, 26343.43022704,  9430.91152033,  1800.73500777,
13261.42480211,  1120.17810533, 10386.66427709, 10567.29006474,
16944.25995713, 26846.54662457,  6939.11178393,  5193.04710054,
 5846.00017265, 13229.60536846, 11098.33930228,  8362.28134289,
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 4157.01930317, 28917.86562624,  -914.37342357,  2873.71150671,
11046.2540774,  15683.06950225,  5210.67532324,  6888.38518351
```
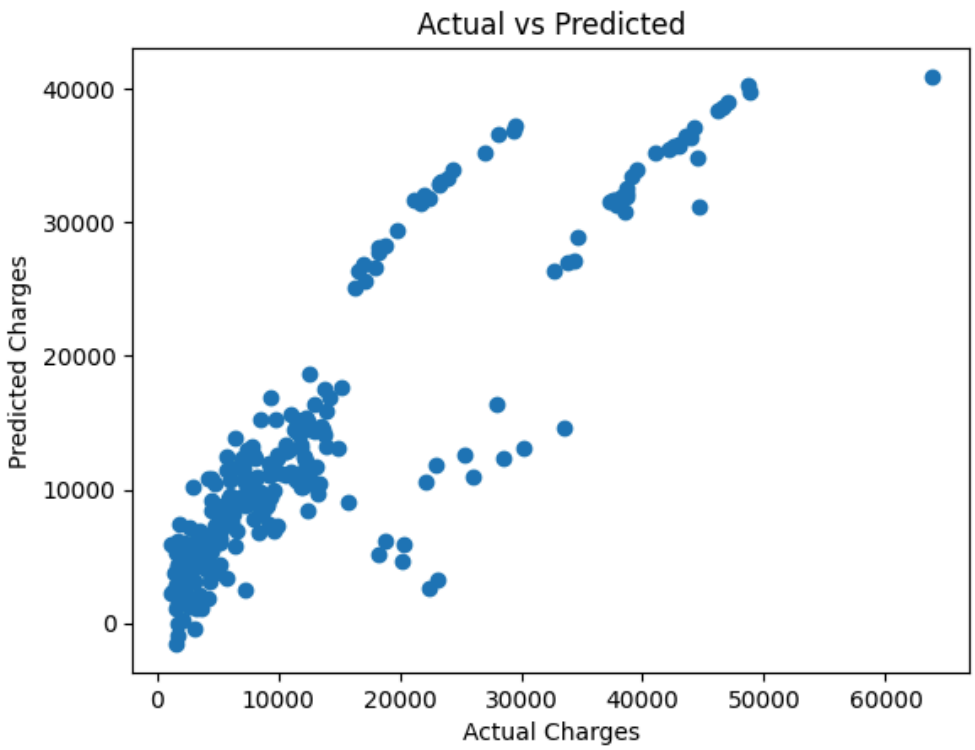

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = mse**0.5 # Calculate RMSE from MSE
r2 = r2_score(y_test, y_pred)

print(f"Mean Absolute Error (MAE): {mae:.2f}")
print(f"Mean Squared Error (MSE): {mse:.2f}")
print(f"Root Mean Squared Error (RMSE): {rmse:.2f}")
print(f"R-squared (R2): {r2:.2f}")
```

```
Mean Absolute Error (MAE): 4181.19
Mean Squared Error (MSE): 33596915.85
Root Mean Squared Error (RMSE): 5796.28
R-squared (R2): 0.78
```

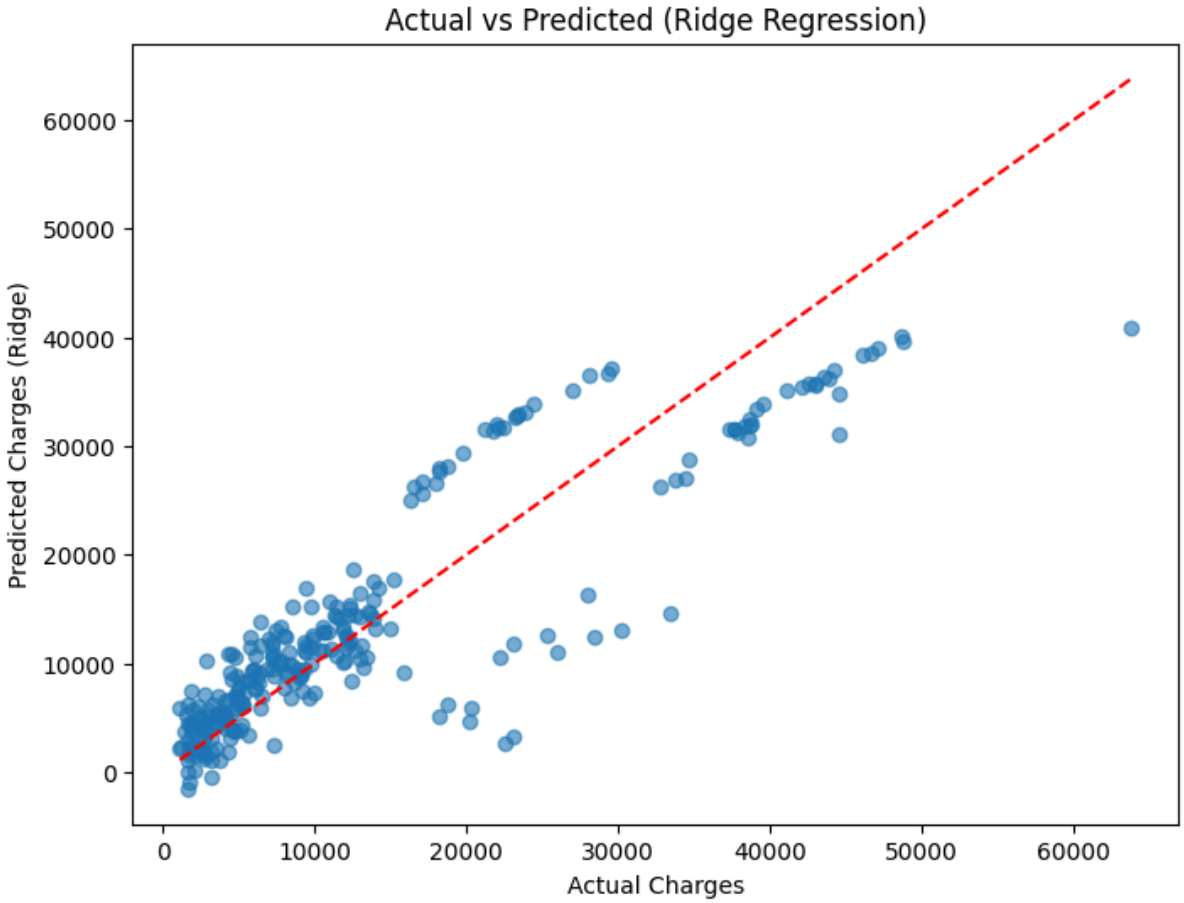
```
import matplotlib.pyplot as plt

plt.scatter(y_test, y_pred)
plt.xlabel("Actual Charges")
plt.ylabel("Predicted Charges")
plt.title("Actual vs Predicted")
plt.show()
```



Trying Ridge Regression

```
#Using Ridge to see how model fits

from sklearn.linear_model import Ridge

# Initialize and train the Ridge model
ridge_model = Ridge(alpha=1.0) # alpha is the regularization strength
ridge_model.fit(X_train, y_train)

# Make predictions
y_pred_ridge = ridge_model.predict(X_test)

# Evaluate the Ridge model
mae_ridge = mean_absolute_error(y_test, y_pred_ridge)
mse_ridge = mean_squared_error(y_test, y_pred_ridge)
rmse_ridge = mse_ridge**0.5
r2_ridge = r2_score(y_test, y_pred_ridge)

print(f"Ridge Model - Mean Absolute Error (MAE): {mae_ridge:.2f}")
print(f"Ridge Model - Mean Squared Error (MSE): {mse_ridge:.2f}")
print(f"Ridge Model - Root Mean Squared Error (RMSE): {rmse_ridge:.2f}")
print(f"Ridge Model - R-squared (R2): {r2_ridge:.2f}")

# Plot actual vs predicted for Ridge
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred_ridge, alpha=0.6)
plt.xlabel("Actual Charges")
plt.ylabel("Predicted Charges (Ridge)")
plt.title("Actual vs Predicted (Ridge Regression)")
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--') # Ideal line
plt.show()
```

```
Ridge Model - Mean Absolute Error (MAE): 4193.59
Ridge Model - Mean Squared Error (MSE): 33645037.09
Ridge Model - Root Mean Squared Error (RMSE): 5800.43
Ridge Model - R-squared (R2): 0.78
```

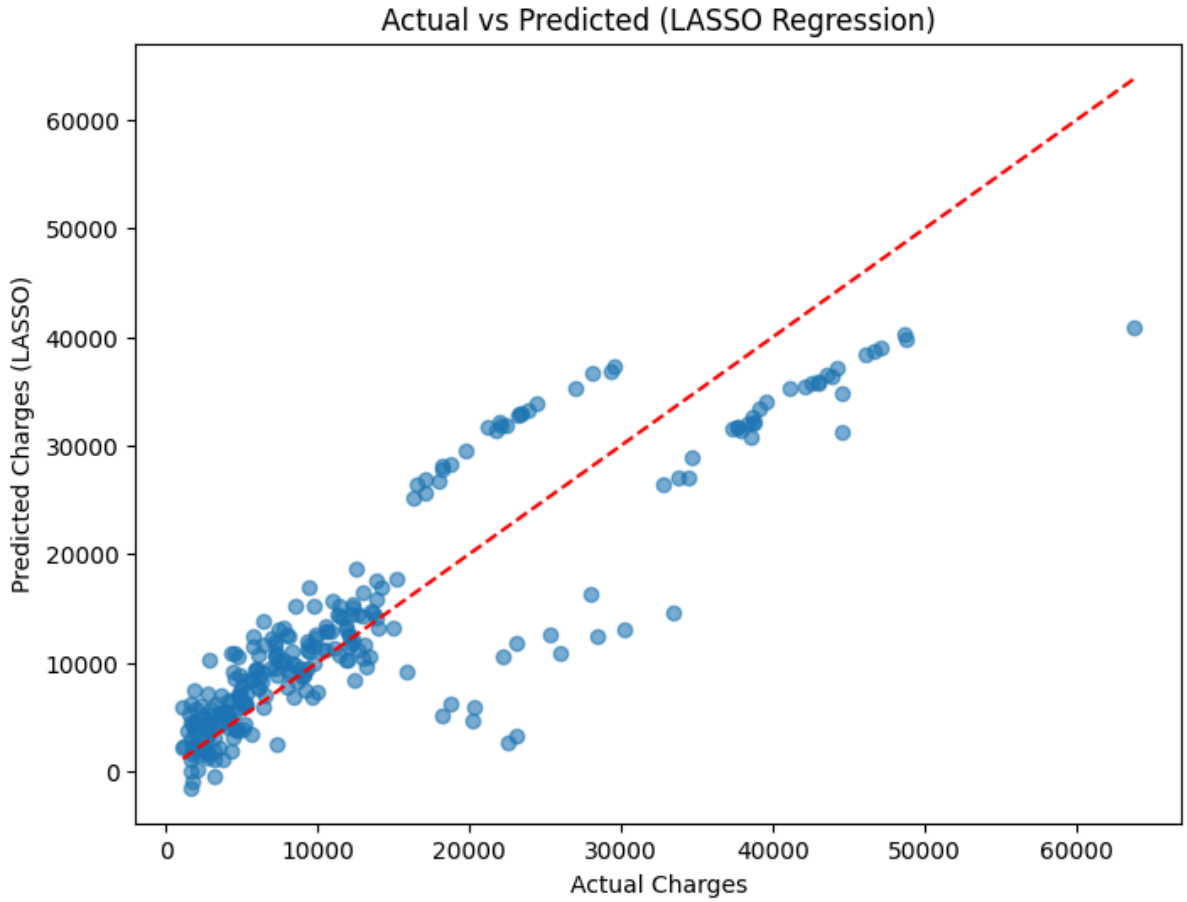


Trying LASSO Regression

```
#Use LASSO to see how the model fits

from sklearn.linear_model import Lasso

# Initialize and train the LASSO model
lasso_model = Lasso(alpha=1.0) # alpha is the regularization strength
lasso_model.fit(X_train, y_train)

# Make predictions
y_pred_lasso = lasso_model.predict(X_test)

# Evaluate the LASSO model
mae_lasso = mean_absolute_error(y_test, y_pred_lasso)
mse_lasso = mean_squared_error(y_test, y_pred_lasso)
rmse_lasso = mse_lasso**0.5
r2_lasso = r2_score(y_test, y_pred_lasso)

print(f"LASSO Model - Mean Absolute Error (MAE): {mae_lasso:.2f}")
print(f"LASSO Model - Mean Squared Error (MSE): {mse_lasso:.2f}")
print(f"LASSO Model - Root Mean Squared Error (RMSE): {rmse_lasso:.2f}")
print(f"LASSO Model - R-squared (R2): {r2_lasso:.2f}")

# Plot actual vs predicted for LASSO
```

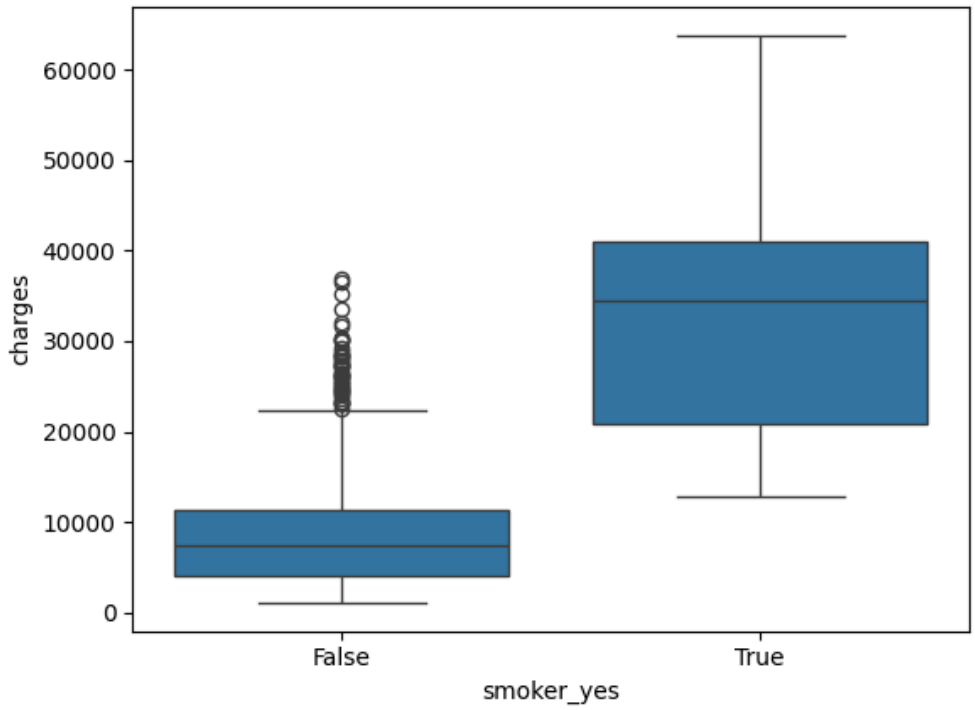
```
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred_lasso, alpha=0.6)
plt.xlabel("Actual Charges")
plt.ylabel("Predicted Charges (LASSO)")
plt.title("Actual vs Predicted (LASSO Regression)")
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--') # Ideal line
plt.show()
```

```
LASSO Model - Mean Absolute Error (MAE): 4182.43
LASSO Model - Mean Squared Error (MSE): 33605507.55
LASSO Model - Root Mean Squared Error (RMSE): 5797.03
LASSO Model - R-squared (R2): 0.78
```




```
#See if smokers costs are higher than nonsmokers

import seaborn as sns

sns.boxplot(x="smoker_yes", y="charges", data=df)
```

```
<Axes: xlabel='smoker_yes', ylabel='charges'>
```



You can compare the metrics and plots from these models to your initial Linear Regression model. Keep in